AIRLINE PASSENGER SATISFACTION

Executive Dashboard

Demographics

Travel & Flight

Service Quality

Customer Satisfaction

Data Mining

OLAP Analysis

EXECUTIVE DASHBOARD

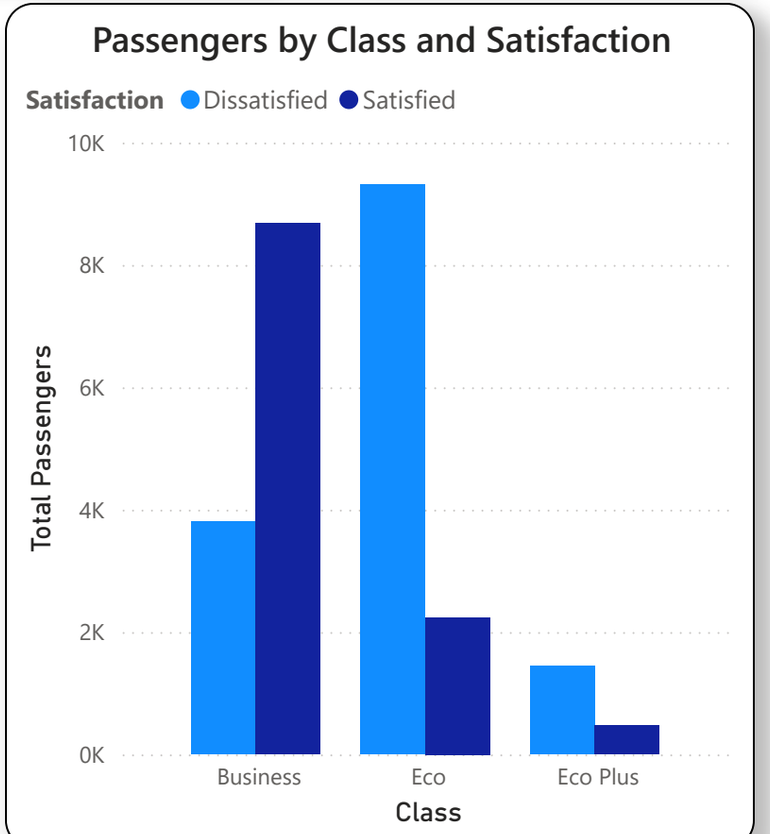
Total Passengers	Satisfied Passengers	Satisfaction Rate	Average Delay
26K	11K	43.91%	14.70

✓ Delay Under Control

Delay Alert

✓ Service Quality Good

Service Alert



Seat_Comfort
- 1
- 2

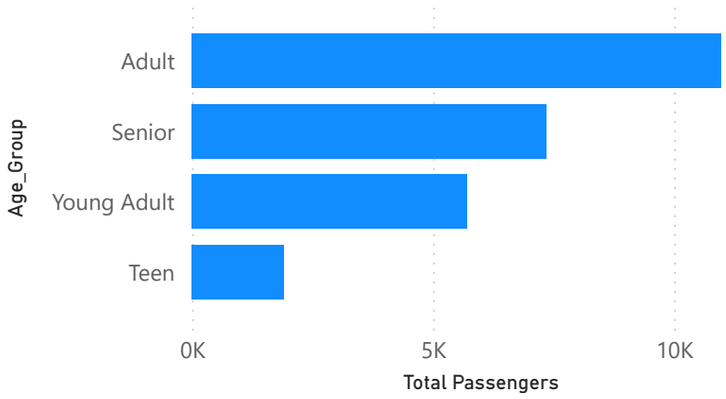
Type_of_Travel
- Business travel
- Personal Travel

Age_Group
- Adult
- Senior
- Teen
- Young Adult

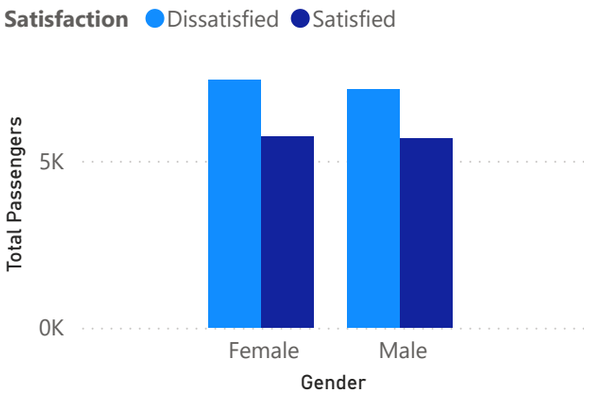
AIRLINE PASSENGER SATISFACTION

PASSENGER DEMOGRAPHICS

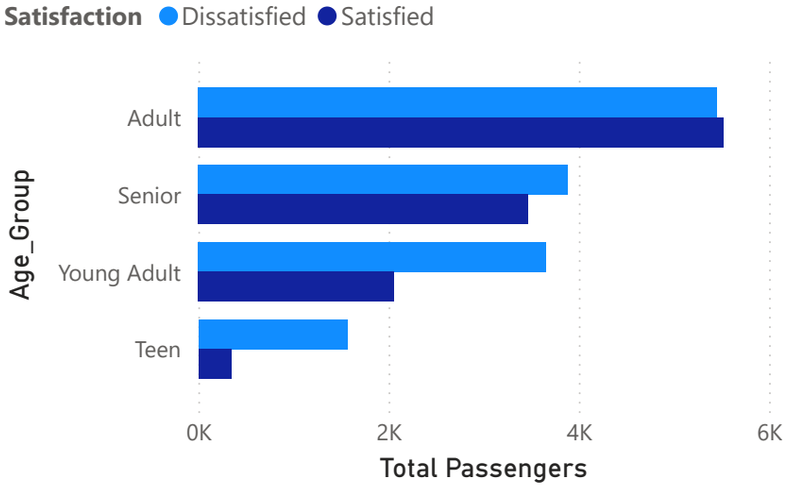
Passenger Distribution by Age Group



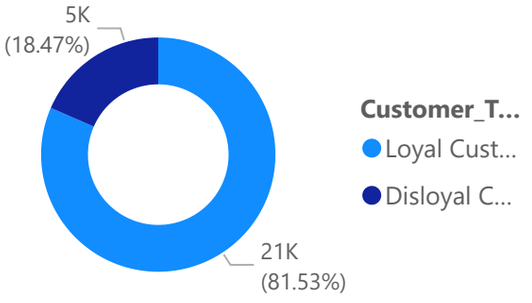
Gender vs Passenger Satisfaction



Age Group vs Satisfaction



Loyal vs Disloyal Customers



Age

- 7
- 8
- 9
- 10

Class

- Business
- Eco
- Eco Plus

Type_of_Travel

- Business travel
- Personal Travel

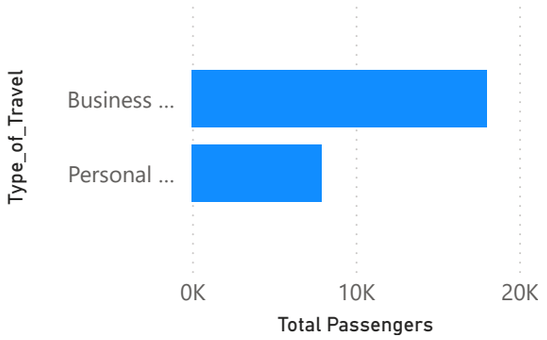
Customer_Type

- Disloyal Custo...
- Loyal Customer

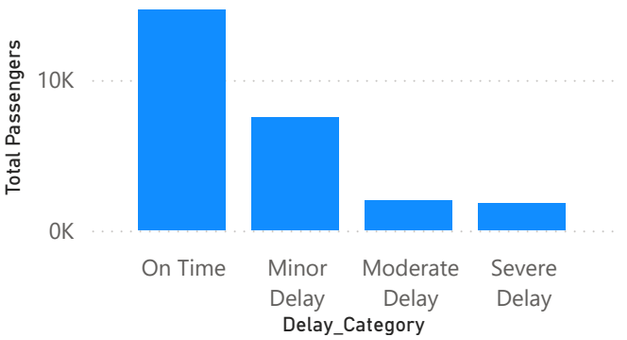
AIRLINE PASSENGER SATISFACTION

TRAVEL & FLIGHT ANALYSIS

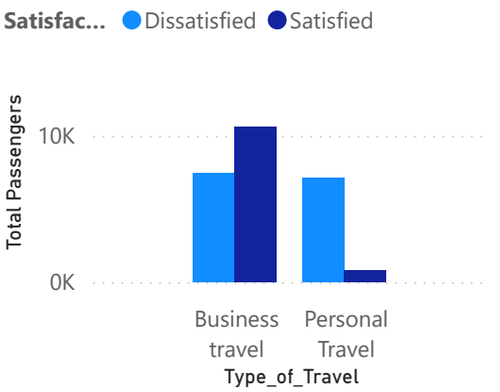
Passenger Distribution by Travel Type



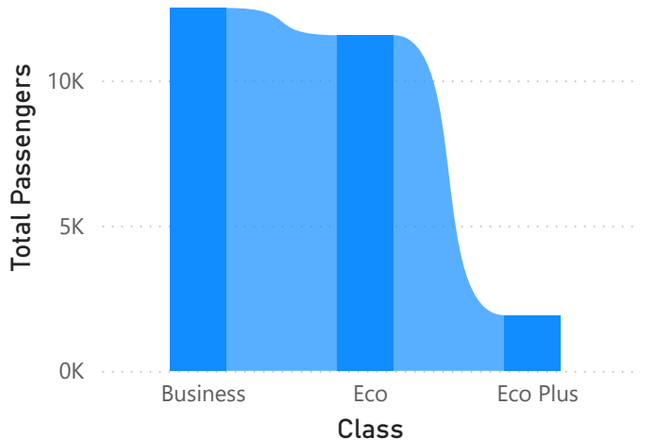
Passenger Distribution by Delay Category



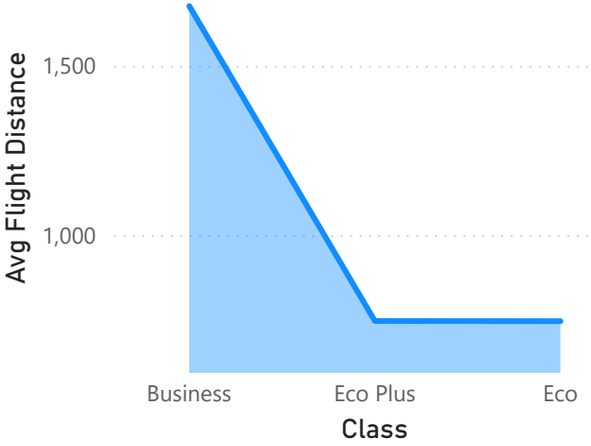
Travel Type vs Satisfaction



Passengers by Travel Class



Average Flight Distance by Class



Class

- [ ] Business
- [ ] Eco
- [ ] Eco Plus

Age_Group

- [ ] Adult
- [ ] Senior
- [ ] Teen
- [ ] Young Adult

AIRLINE PASSENGER SATISFACTION

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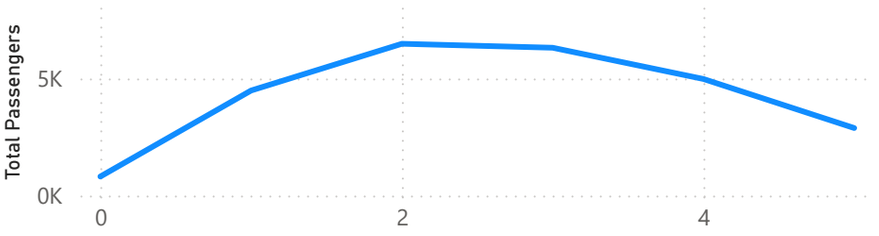
SERVICE QUALITY ANALYSIS

Seat_Comfort	0	1	2	3	4	5	Total
1		2260	208	188	200	174	**3030**
2		243	2641	295	278	175	**3632**
3		445	517	2961	520	187	**4630**
5		119	178	1177	1236	3976	**6686**
4	2	341	437	1442	4552	1213	**7987**
Total	**2**	**3408**	**3981**	**6063**	**6786**	**5725**	**25965**

Passenger Distribution by Wifi Service Rating

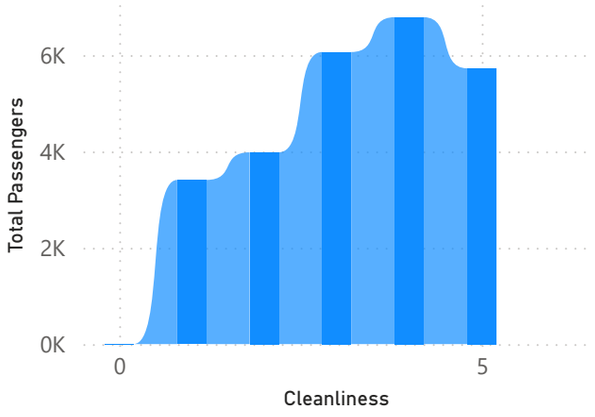


Satisfaction
- Dissatisfied
- Satisfied

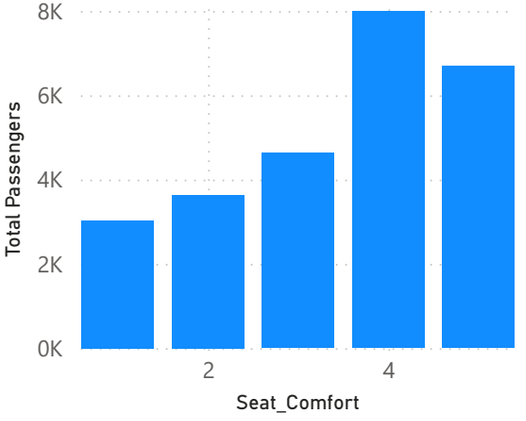
Type_of_Travel
- Business travel
- Personal Travel

Age_Group
- Adult
- Senior

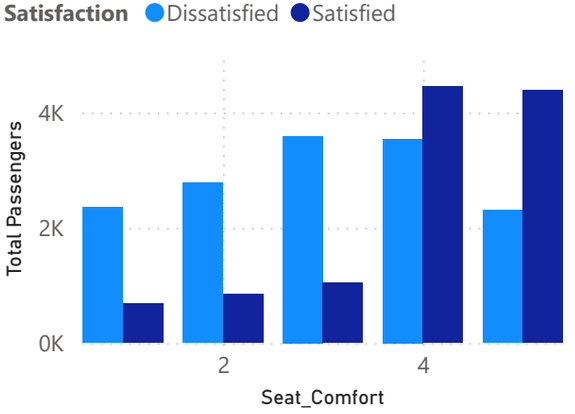
Passenger Distribution by Cleanliness Rating



Passenger Distribution by Seat Comfort Rating



Seat Comfort Rating vs Passenger Satisfaction



AIRLINE PASSENGER SATISFACTION

Executive Dashboard

Demographics

Travel & Flight

Service Quality

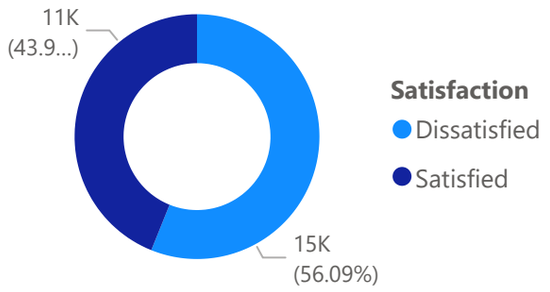
Customer Satisfaction

Data Mining

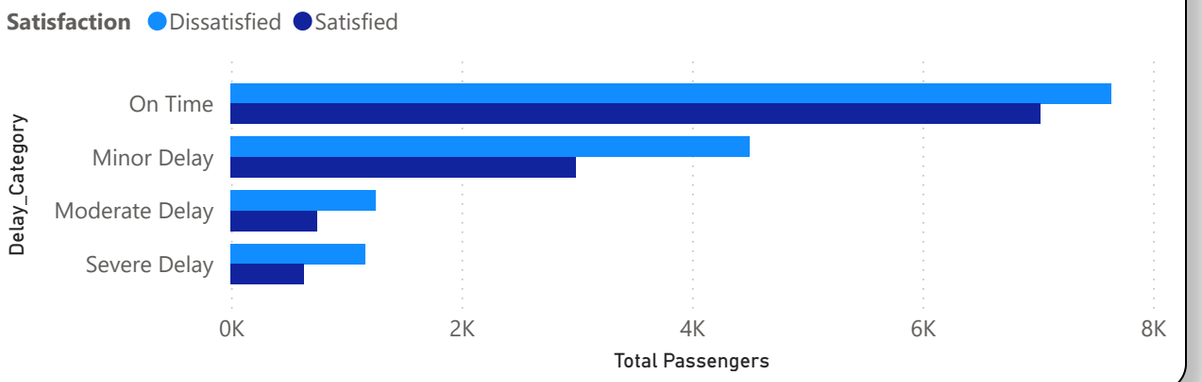
OLAP Analysis

CUSTOMER SATISFACTION ANALYSIS

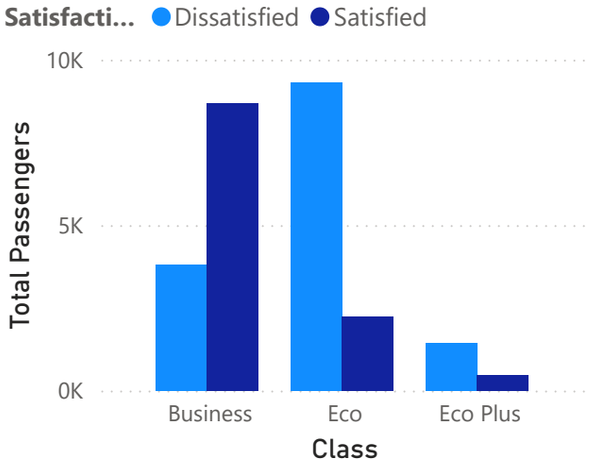
Overall Passenger Satisfaction



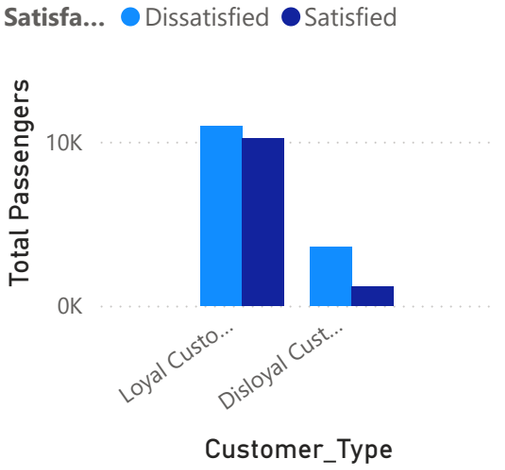
Flight Delay Impact on Satisfaction



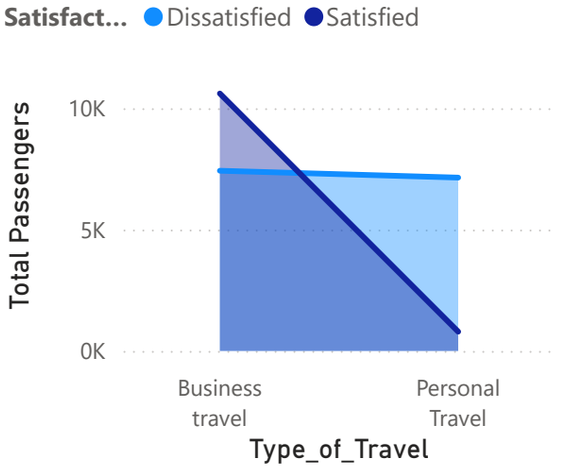
Travel Class vs Passenger Satisfaction



Customer Loyalty vs Satisfaction

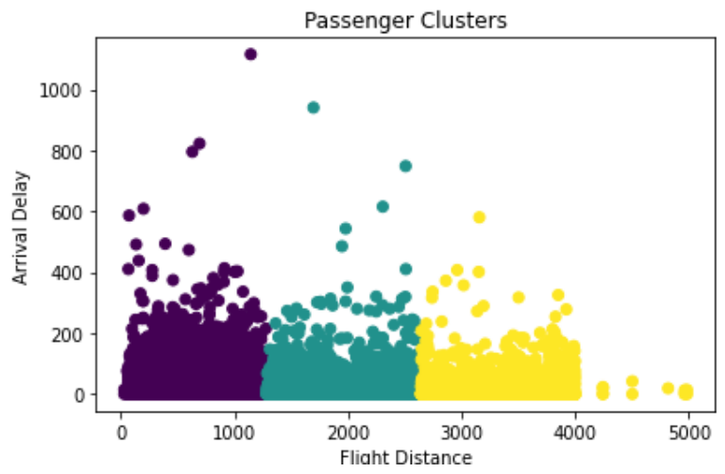


Travel Type vs Passenger Satisfaction

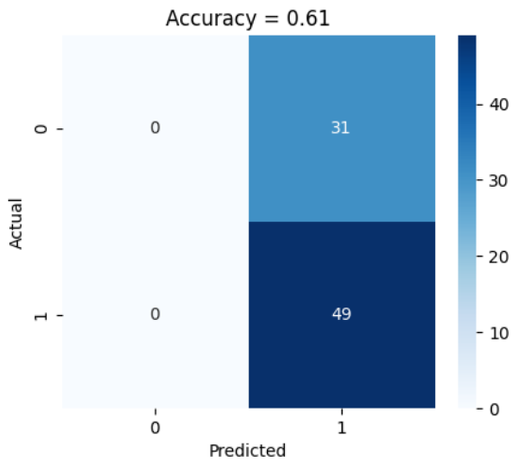


PREDICTIVE & DATA MINING

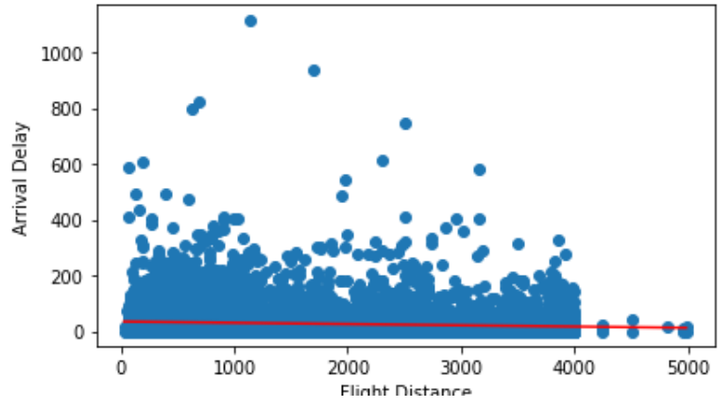
Clustering (K-Means)



Logistic Regression Classification (Satisfaction)



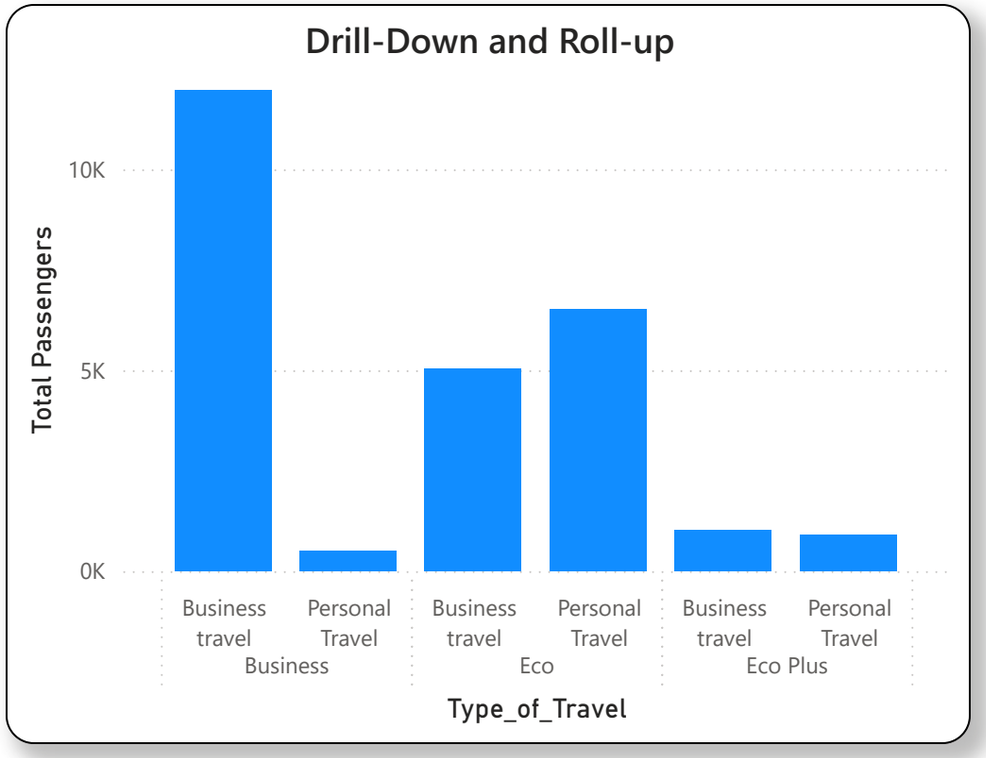
Linear Regression (Distance vs Delay)



Key Insights:

- Passenger clusters reveal travel behavior patterns.
- Logistic regression predicts satisfaction with ~61% accuracy.
- Flight distance shows weak correlation with arrival delay.

OLAP ANALYSIS



Class	Dissatisfied	Satisfied	Total
Business	3807	8685	**12492**
Eco	9316	2240	**11556**
Eco Plus	1442	475	**1917**
Total	**14565**	**11400**	**25965**

Slicer (Customer Type)

- [ ] Disloyal Customer
- [ ] Loyal Customer

Dice (Seat ...

- [ ] 1
- [ ] 2

Dice (Delay Category)

- [ ] Minor Delay
- [ ] Moderate Delay

- OLAP operations enable multidimensional exploration of airline passenger data.
- Drill-down and roll-up reveal passenger distribution across travel hierarchies.
- Slice filters the data cube on a single dimension such as customer type.
- Dice applies multiple filters simultaneously.
- Pivot reorganizes the data to analyze passenger satisfaction across travel classes.